

STIX-based Network Security Knowledge Graph Ontology Modeling Method

Zhengjun Liu, Zhi Sun, Jianfeng Chen
China Electronic Technology Corporation Research
Institute Co., Ltd Xiongan, China
Cyberspace Security Key Laboratory of Sichuan
Province
China Electronic Technology Cyber Security Co., Ltd,
Chengdu, China
james_liuzj@sina.com,
sunzhi1019@163.com,
atrix@163.com

Yujin Zhou, Tao Yang, Hui Yang, Jie Liu
China Electronic Technology Corporation Research
Institute Co., Ltd Xiongan, China
China West Normal University, Nanchong, China
China Electronic Technology Cyber Security Co., Ltd,
Chengdu, China
zhouyujinanna@126.com,
yangtao_cwnu@163.com,
yanghui032178@163.com,
liujie.bupt@gmail.com

ABSTRACT

Network security incidents are complex and unstructured, making them difficult to understand and share. In this paper, we analyze the commonality between structured threat information representation (STIX) and network security domain knowledge, and propose a knowledge graph ontology modeling method of network security based on STIX. With the architecture knowledge of STIX, this method generates an ontology schema of network security knowledge graph, through classifying the concepts in the field of network security, describing the attributes of concepts and combining the relationships between concepts. The ontology schema has small redundancy and strong structural hierarchy, and can clearly display the structure of the attack activity and the mutual relationship. Therefore, it can help decision makers to understand security incidents more deeply, and help them make reasonable decisions and share cyber threat intelligence.

CCS Concepts

• Information systems→Data management systems→Database design and models→Graph-based database models

Keywords

Knowledge Graph; Ontology; STIX; Network Security

1. INTRODUCTION

Due to the vulnerability of the Internet, the number of cyber threats has increased significantly, and the attack surface and the complexity of network attacks have also expanded rapidly. For example, Advanced Persistent Threat (APT) consists of thousands of attacks, with millions of victims and a time span of several years. It is difficult for cyber security decision makers to clarify

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.

ICGDA 2020, April 15–17, 2020, Marseille, France

© 2020 Association for Computing Machinery.

ACM ISBN 978-1-4503-7741-6/20/04...\$15.00

DOI: <https://doi.org/10.1145/3397056.3397083>

the intrinsic link between attacks, and to make reasonable decisions to prevent or reduce the damage of attacks.

Knowledge graph use visualization technology to deeply display and describe the relationship between attack organization, attack control resources, attack methods, attack targets and attack events in network security events. It provides an effective basis for decision makers to make accurate defenses, and helps prevent or reduce the damage of various types of attacks.

The ontology schema is used to describe the concepts and their connections in a wide range or within a certain domain, so that these concepts and connections have a clear and unique definition within the scope of sharing. The construction of ontology schema is an important part of the knowledge graph construction. The quality of the ontology schema determines whether the constructed knowledge graph structure is clear and the logic is reasonable. Researchers at home and abroad have conducted in-depth research on the methods of ontology modeling.

Noy[1], the professor of Stanford, first proposed the ontology modeling approach, which manually built the ontology schema in seven steps. Che[2] of Northeastern University summarized the method of constructing the ontology schema from structured data in databases, XML documents and Web forms. A significant work was completed by Undercoffer et al. [3] from the University of Maryland, who developed an ontology schema to model attacks and related entities. Their ontology is only targeted at attacks and has certain limitations. In order to represent concepts and entities related to the field of network security, Joshi et al. [4] proposed the network security ontology derived from the ontology proposed by Undercoffer. They extended the ontology to a model relationship that captures the US National Vulnerability Database (NVD) pattern structure and the concept of security vulnerabilities. This ontology contains 11 types of entities (e.g., vulnerabilities, products, methods, consequences, etc.). More et al[5]. also extended the ontology proposed by Undercoffer and added rules to the reasoning logic. Their ontology schema consists of three basic categories: methods, consequences, and targets.

In addition, a corporation named MITRE is also studying how to develop ontology for the network security domain [6, 7]. MITRE has created multiple standards and datasets for specific topic areas of the network security domain, and they have a rich database. Based on the efforts of Undercoffer and MITRE, Iannacone et al.

[8] proposed the ontology of the network security knowledge base. This ontology represents the result of an iterative design process, which aims to create a representation of knowledge that can effectively merge data from different data sources within the cyberspace security domain. The resulting ontology contains 15 entity types and 115 properties in total. Recently, Jia et al. [9] proposed a practical approach for constructing a network security knowledge graph based on a quintuple model. Using machine learning, this method extracted entities and built ontology schema to obtain a network security knowledge base. Chen et al. [10] proposed a ontological threat intelligence sharing in cyberspace Security, which can correctly express the threat characteristics, threat activities, security incidents and other conceptual features contained in threat intelligence. It provided a feasible solution for understanding and communicating information between multiple intelligence individuals.

Structured Threat Information Expression (STIX) is a language and serialization format used to exchange cyber threat intelligence. It can effectively display the relationship between the various elements of the attack event. However, there are currently few studies on ontology modeling based on STIX. In this paper, we propose a STIX based network security knowledge graph ontology modeling method. We analyze the commonality between STIX and network security, extract key classes in the field of network security and analyzes the relationship between classes. Then, we construct the ontology schema of network security knowledge graph based the analysis.

2. STIX

No organization has enough relevant network intelligence knowledge to threaten situational awareness, so threat intelligence needs to be shared through trusted partners and community organizations. Through information sharing, partners can get relatively comprehensive and rich contextual intelligence data for better network security defense. For example, Organization A has been attacked, but Organization B has not found any signs of the attack. If Organization A shares the relevant information about the attack, Organization B can use relevant information to discover, detect, and defend this attack. At present, the defense system is built on various security products. Different products need to use shared information automatically, so standardized and structured shared information formats are needed, so that each product can automatically use intelligence.

Structured Threat Information Expression (STIX) is a language and serialization format used to exchange cyber threat intelligence (CTI), and it is one of the most widely used threat intelligence sharing languages. The US Department of Homeland Security (DHS) uses STIX through TAXII, and distributes threat intelligence to partners through STIX. National Cyber Security and Communications Integration Center of DHS is gradually implementing the production of all network threat related operational products based on STIX. In July 2013, Microsoft announced that they will use STIX to share threat information through the new MAPP for Responders program. In September 2013, HP announced the launch of a new threat centric threat intelligence sharing environment based on STIX. In 2012, the Japan Financial Services Information Sharing and Analysis Center (FS-ISAC) agreed to implement the STIX architecture (including CybOX) to enable cyber threat information sharing among members of the financial services sector. As of May 2013, FS-ISAC used STIX to create an operational threat information base for use by all 4,200 member organizations. Japan Information Technology Promotion Agency is currently conducting a positive

feasibility study to apply the elements of the STIX architecture (CybOX, MAEC, etc.) as an international exchange format for network observables and threat information.

With STIX, all aspects of suspicion, compromise and attribution can be represented clearly with objects and descriptive relationships. Contributing and ingesting CTI becomes a lot easier. STIX2 defines twelve STIX Domain Objects (SDOs): *Attack Pattern*, *Campaign*, *Course of Action*, *Identity*, *Indicator*, *Intrusion Set*, *Malware*, *Observed Data*, *Report*, *Threat Actor*, *Tool* and *Vulnerability*. STIX information can be visually represented for an analyst or stored as JSON to be quickly machine readable. The following is a JSON-based example of a STIX 2.0 Campaign object

```
{
  "type": "campaign",
  "id": "campaign--8e2e2d2b-17d4-4cbf-938f-98ee46b3cd3f",
  "created": "2016-04-06T20:03:00.000Z",
  "name": "Green Group Attacks Against Finance",
  "description": "Campaign by Green Group against targets in the
    financial services sector."
}
```

STIX2 defines two STIX Relationship Objects (SROs): Relationship is used to link two SDOs and to describe how they are related to each other. Sighting is used to denote the belief that an element of CTI was seen (e.g., indicator, malware).

3. THE MODELING METHOD

Knowledge graph is divided into data layer and ontology schema. The data layer is composed of a series of fact, and the ontology schema is used to construct entities, attributes, and relationships from the data. Ontology schema is the core of the knowledge graph, and it determines the quality of the constructed knowledge graph. In this section, we will describe the proposed method based on STIX.

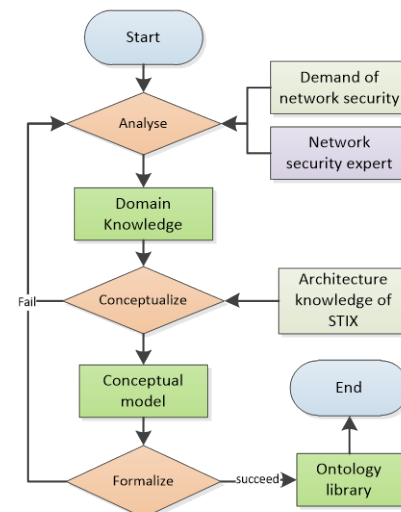


Fig.1 The process of STIX-based network security knowledge graph ontology modeling.

Fig.1 shows the process of network security knowledge graph ontology modeling based on STIX. The process consists of three main stages, as show in the diamond area. The first stage is the

domain analysis. In the analysis phase, the experts in the field of network security combine the needs of constructing a knowledge graph to determine the requirements of the ontology schema. For example, what is the specific scope and use of ontology, what information needs to be described in the field of network security, what questions to answer, and who will use and maintain the ontology library, etc. The second stage is the conceptualization, which is the most important stage in ontology modeling. According to the domain knowledge obtained in the previous stage, we combine with STIX's architectural knowledge to build a conceptual model, which defines the main classes and their inheritances, and defines attributes and relationships. The final stage is formalization. At this stage, the conceptual model is formalized into a qualified ontology. By adding constraints of relationships and attributes, the quaternion form of the ontology is formed and added into ontology library. We will introduce the conceptualization and formalization in the process of ontology modeling in detail later.

3.1 Conceptual Modeling

3.1.1 Class and attribute analysis

Through the analysis of the requirements of the network security field, the network security knowledge graph needs to describe the scenario that some attackers use a number of tools to perform a series of attacks on the vulnerabilities exposed by certain cyberspace resources, and these attacks can be observed with some characterization information and be mitigated by some course of actions. Fig.2 shows the scene of the attack activity. The decision maker can use the knowledge graph to query related information and visually display the query results. Reasoning based on knowledge graph enables decision makers to make better decisions.

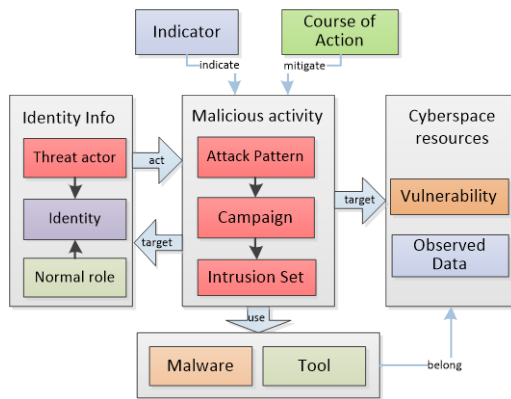


Fig.2 The scene description of network security events.

In this chapter, we first identify the main classes that need to be modeled in the field of network security. As show in Fig.3, there are thirteen classes in our model.

Identity, *Threat Actor* and *Normal role* are three classes that define the identity of a real person or group. *Identity* describes the detailed characteristics of a person or group. *Threat Actor* and *Normal role* can be projections (subclass) of *Identity*, or they can exist independently of any *Identity*. An entity of *Threat Actor* can attack that of *Normal role* or *Identity*.

Attack Pattern, *Campaign* and *Intrusion Set* are three different levels of malicious activities, where *Attack Pattern* is the basic, fixed attack method, *Campaign* is a short-term attack campaign,

including one or more *Attack Patterns*, and the *Intrusion Set* has a longer time span and may contain one or more *Campaigns*.

Malware, *Tool* and *Cyberspace resources* are the basic resource classes. *Malware*, *Tool* is the subclass of *Cyberspace resources*, which is used by an attacker in the attack activity. We separate them as an independent class to better describe cyber security threat intelligence.

Observed Data, *Indicator* and *Vulnerability*, are observation classes. Among them, *Observed Data* is a general observation value with no characterization, and *Indicator* is an observation that can represent the uniqueness of the object. *Vulnerability* is the vulnerability of cyberspace resources and is the target of attack activity. There is one action class in our model, named *Course of Action*, which can mitigate the behavior of threats.

The characteristics and related attributes of each class are described in detail below:

Identity can represent actual individuals, organizations, or groups, as well as classes of individuals, organizations, or groups. The Identity object can capture basic identifying information, contact information, label, and the sectors that the Identity belongs to. The label of identities includes individual, group, organization, class (such as all hospitals, all Europeans, and the leadership of a system) and unknown. The sectors contain agriculture, aerospace, automotive, communications, construction, defence, education, energy, entertainment, financial-services, government-national, government-regional, government-local, government-public-services, healthcare, hospitality-leisure, infrastructure, insurance, manufacturing, mining, non-profit, pharmaceuticals, retail, technology, telecommunications, transportation, utilities, etc.

Threat Actors are actual individuals, groups, or organizations believed to be operating with malicious intent. Threat Actors can be characterized by their motives, labels, capabilities, goals, sophistication level, past activities, resources they have access to, and their role in the organization. The labels of threat actors contain activist, competitor, crime-syndicate, criminal, hacker, insider-accidental, insider-disgruntled, nation-state, sensationalist, spy, terrorist. The roles of threat actors are agent, director, independent, infrastructure-architect, infrastructure-operator, malware-author and sponsor. The sophistication level of threat actor are none, minimal, intermediate, advanced, expert, innovator and strategic.

Normal roles are individuals, groups, or organizations who performs normal activities in cyberspace. The information of a normal role is less than that of a threat actor, and a normal role can change to a threat actor at any time. Normal role can be characterized by ID, name, description, label, role, etc.

Attack Pattern is a type of Tactics, Techniques, and Procedures (TTP) that describes ways threat actors attempt to compromise targets. According to the implementation mechanism of Attack Pattern, it is divided into 9 categories which contain 517 attacks. These categories are Engage in Deceptive Interactions, Abuse Existing Functionality, Manipulate Data Structures, Manipulate System Resources, Inject Unexpected Items, Employ Probabilistic Techniques, Manipulate Timing and State, Collect and Analyze Information, Subvert Access Control. Each attack contains properties which named ID, Name, Status, Description, Typical Severity Execution Flow, Related Weaknesses. In this paper, nine attack pattern categories are marked as knowledge graph type, and sub-knowledge graph of attack pattern is sequentially established. This sub-knowledge graph does not increase with the increase of

network security events, but only expands with the discovery of new attack patterns. For specific network security events, we add a subclass of attack pattern called event attack mode. Its properties are ID, name, CAPEC-ID, created_time, modified_time, kill_chain_phases, description, where CAPEC-Id is the ID attack pattern.

Campaign is a grouping of adversarial behaviors that describes a set of malicious activities or attacks that occur over a period of time against a specific set of targets. Campaigns usually have well defined objectives and may be part of an Intrusion Set. Campaigns can be characterized by their objectives and the incidents they cause, people or resources they target, and the resources (infrastructure, intelligence, Malware, Tools, etc.) they use. Its properties are name, description, aliases, first_seen, last_seen, objective.

Intrusion Set is a grouped set of adversarial behaviors and resources with common properties believed to be orchestrated by a single threat actor. An Intrusion Set may capture multiple Campaigns or other activities that are all tied together by shared attributes indicating a common known or unknown Threat Actor. Where a Campaign is a set of attacks over a period of time against a specific set of targets to achieve some objective, an Intrusion Set is the entire attack package and may be used over a very long period of time in multiple Campaigns to achieve potentially multiple purposes. Its properties are name, description, aliases, first_seen, last_seen, goals, resource_level, primary_motivation, secondary_motivations. The resource level of attacks contains individual, club, contest, team, organization, government. The motivation of attacks contains accidental, coercion, dominance, ideology, notoriety, organizational-gain, personal-gain, personal-satisfaction, revenge, unpredictable.

Cyberspace resource is the general term for all hardware devices and all virtual resource (software and information) in the field of cyberspace. Network hardware devices include hosts, servers, routers, switches, probes, etc. Virtual resources include databases, operating systems, software systems, IP, mailboxes, files, virtual

currency, etc. malware and tool is one kind of virtual resource. Vulnerability and Observed Data is the feature of some cyberspace resources.

Malware is a type of TTP, also known as malicious code and malicious software, used to compromise the confidentiality, integrity, or availability of a victim's data or system. The properties of malware are ID, name, description, category_label, kill_chain_phases. The category label of malware are adware, backdoor, bot, ddos, dropper, exploit-kit, keylogger, ransomware, remote-access-trojan, resource-exploitation, rogue-security-software, rootkit, screen-capture, spyware, trojan, virus, worm.

Tool is legitimate software that can be used by threat actors to perform attacks. Remote access tools (e.g., RDP) and network scanning tools (e.g., Nmap) are examples of Tools that may be used by a Threat Actor during an attack. It contains properties to name and describe the tool, a list of Kill Chain Phases the tool can be used to carry out, the version of the tool, and the label of the tool. The labels of the tool contain denial-of-service, exploitation, information-gathering, network-capture, credential-exploitation, remote-access, vulnerability-scanning, which can be used as the type label of knowledge graph.

Vulnerability is a mistake in software that can be directly used by a hacker to gain access to a system or network. Vulnerability is primarily used to link to external definitions of vulnerabilities or to describe 0-day vulnerabilities that do not yet have an external definition. It can be characterized by name, ID, description, create_time and external_references. The external_references may be one or more external vulnerability, such as CVE ID.

Observed Data Conveys information observed on a system or network (e.g. an IP address) using some Cyber Observable specification. Observed Data can capture the observation of an IP address, a network connection, a file, or a registry key. Observed Data is not an intelligence assertion, it is simply information: this file was seen, without any context for what it means. It can be characterized by ID, name, create_time, first_observed, last_observed, number_observed, objects.

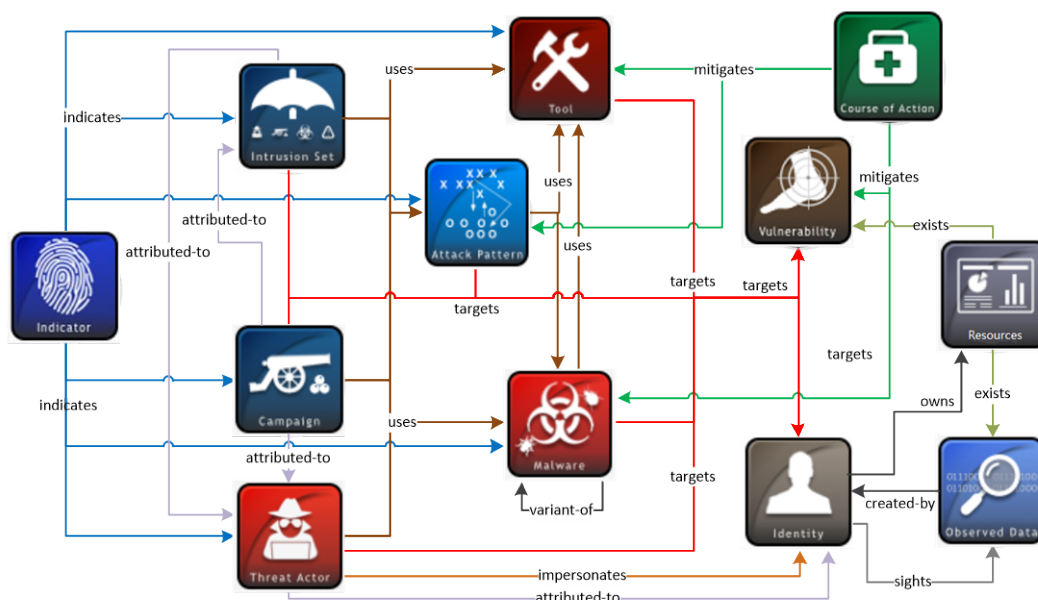


Fig.3 The relationship of classes.

Indicator contains a pattern that can be used to detect suspicious or malicious cyber activity. The Indicator contains a simple textual description, a label that specifies the type of it, the Kill Chain Phases that it detects behavior in, a time window for when the Indicator is valid or useful, and a required pattern property to capture a structured detection pattern. The label of Indicator includes anomalous-activity, anonymization, benign, attribution, compromised, malicious-activity, etc.

Course of Action is an action taken to either prevent an attack or respond to an attack that is in progress. It may describe technical, automatable responses (applying patches, reconfiguring firewalls) but can also describe higher level actions like employee training or policy changes. For example, a course of action to mitigate a vulnerability could describe applying the patch that fixes it.

Cyberspace Threat Intelligence(CTI) is a collection of one or several objects described above. It contains all the elements of an attack activity, and can be characterized by its name, ID, create-time, number_of_objects.

3.1.2 Inter-class relationship analysis

In the knowledge graph of network security, the main types of inter-class relationship are *Indicates, Uses, Attributed-to, Targets, Mitigates, Impersonates, Sights, Owns, Exists, Create, Variant-of, belong*, etc. Fig.3 shows the main relationship of the classes described in the above chapter.

Moreover, there is a relationship between CTI and other classes, named *bundle*. This relationship means that one object of CTI has one or several objects of other classes.

3.2 Formalization

In this section, we will formalize the conceptual model. First, we need to determine the common types of properties and their associated constraints. Table.1 shows the type we defined in our ontology schema. The properties include the value property of class and the relationship property between classes. Types of the property contain Boolean, binary, hex, integer, float, string, timestamp, class_id and open_voc. The first letter of all class names is capitalized. All relationship names, property names and literals are in lowercase. All class names, property names, relationship names, and vocabulary terms are between three and 250 characters long.

Table.1 the type and description of property

Type	Description
boolean	A value of true or false.
binary	A constant of binary type is a base64 encoded array of octets (8-bit bytes) per. The base64 string must be surrounded by apostrophes and prefixed by a 'b'(e.g., b'ABI=).
hex	A constant of hex type encodes an array of octets (8-bit bytes) as hexadecimal. The string must consist of an even number of hexadecimal characters and prefixed by an 'h' (e.g., h'0012').
integer	A constant of integer type encodes a signed decimal number in the usual fashion (e.g., 123). Negative integers must be represented by prepending a hyphen-minus.
float	A constant of float type encodes a floating point number in the usual fashion (e.g., 123.456).

string	A constant of string type encodes a string as a list of Unicode code points surrounded by apostrophes. The escape character is the backslash ('\'). Only the apostrophe or the backslash may follow, and in that case, the respective character is used for the sequence.
timestamp	The timestamp string must be surrounded by apostrophes and prefixed with a 't'(e.g., t'2019-11-13T07:03:17Z').
class_id	The identifier of class, this string must be prefixed with class name and followed by 8 bit date type and 8 bit random hex number(e.g., malware_20191102_f2343b2a)
open_voc	A value from an open or suggested vocabulary, such as industry sector, attack motivation, malware label.

We define the detail type and constraints of each class. Table.2 shows formalization of identity. Due to space limitations, we will not describe the formalization of other classes in detail.

Table.2 the formalization of identity

Name	type	constraints
id	class_id	Id of this identity, such as 'identity_20191102_aad123fd'
name	string	The name for this identity
create_time	timestamp	>t'1900-01-01T00:00:00Z'
type	open_voc	Including individual, group, organization, class, other.
label	open_voc	The list of roles that this Identity performs (e.g., CEO, Domain Administrators, Doctors, Hospital, or Retailer).
description	string	A description that provides more details about the Identity, potentially including its purpose and its key characteristics.
revoke	boolean	Whether this identity is revoked.
address	string	< 250 characters
sectors	open_voc	The list of industry sectors that this Identity belongs to.
contact_information	string	The contact information (e-mail, phone number, etc.)
created_by	class_id	The identity created this identity.
modified_by	class_id	The identity that modified this identity. This relationship has a modified_time property.
sights	class_id	The Observed Data that sighted by this identity.
owns	class_id	The Resource that identity owns.

In the formalization process, we define the *Ontology* tag to represent the class concept, the *Entity* tag to represent a class instance, the *Relationship* tag to represent the relationship, the tag *Property* to define the property of class, the *Literal* tag to define the value of property, and the *subClassOf* to represent the inheritance relationship of the class. We use the quaternion <S, P, O, A> to represent a piece of knowledge. S is the active object, which can be *Ontology* or *Entity*. P represents a property, which

can be *Relationship* or *Property*. O means passive object, which can be *Ontology* or *Entity* or *Literal*. A represents the attribute of the relationship. A is empty when the relationship has no specific attribute. So, a piece of knowledge can represent one of the following formats.

< *Ontology*, *Relationship*, *Ontology*, A > ,

< *Entity*, *Relationship*, *Ontology*, A > ,

< *Entity*, *Relationship*, *Entity*, A > ,

< *Entity*, *Property*, *Literal*, A >

The following is an example of a piece of knowledge in *Jason* format. We can use it to create knowledge graph. As shown in Fig.4, *Cyberspace Resource* consists of *Device*, *Information* and *Software*, and a role in *Group*, *Nation* and *Person* can also be a *Normal role*.

```
{
  "S": {"prefix": "Entity",
        "name": "liu feng",
        "flag": ""},
  "P": {"prefix": "Relationship",
        "name": "owns",
        "flag": "" },
  "O": {"prefix": "Entity",
        "name": "router_110",
        "flag": "" },
  "A": {"since_time": 't'2019-11-10T17:03:17Z' }
}
```

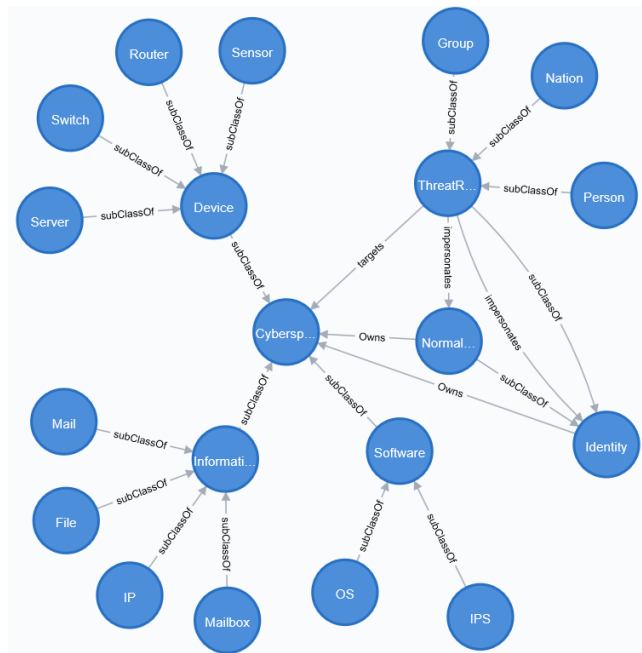


Fig.4 Part of the ontologies in neo4j.

4. CONCLUSIONS AND FUTURE WORK

Network security incidents are intricate and complex, and it is difficult to clarify the relationship between the incidents. This paper draws on the architectural knowledge of STIX, and

proposes a method of network security knowledge graph ontology modeling based on STIX. This method analyzes the conceptual and formal approaches of network security knowledge ontology, and provides a feasible solution for the establishment of network security knowledge graph. It can help the decision makers understand network security incidents clearly and deeply, and make appropriate defense measures.

In future, we will construct the network security knowledge based on our ontology schema. The most important work will be to enrich the cyber security knowledge base and the rules of deduction, and then apply them to the research of intrusion detection and situational awareness.

5. ACKNOWLEDGMENTS

We are grateful for the support of Sichuan Science and Technology Program (No.2017GZDZX0002, No.2019JDRC0084), National Natural Science Foundation of China (No.61803352), Key R & D projects of Sichuan Science and technology plan (No.2018GZ0101), Meritocracy Research Funds of China West Normal University (No.17YC190).

6. REFERENCES

- [1] Noy N, McGuinness D L . Ontology Development 101[J]. Knowledge Systems Laboratory, 2001.
- [2] CHA Songli, MA Zong-min, JIAO Xiao-long. Survey on methodology for constructing ontology based on structured information source. *Application Research of Computers*, 2012;p.12-16.
- [3] Undercoffer J, Joshi A, Pinkston J. Modeling computer attacks: An ontology for intrusion detection. In: Vigna G, Jonsson E, Kruegel C, editors. RAID 2003: Recent advances in intrusion detection. Berlin: Springer; 2003. p. 113–135.
- [4] Joshi A, Lal R, Finin T, Joshi A. Extracting cybersecurity related linked data from text. In: *Proceedings of the 7th IEEE international conference on semantic computing*. Los Alamitos: IEEE Computer Society Press; 2013. p. 252–259.
- [5] More S, Matthews M, Joshi A, Finin T. A knowledge-based approach to intrusion detection modeling. In: *Proceedings of 2012 IEEE symposium on security and privacy workshops*. Los Alamitos: IEEE Computer Society Press; 2012. p.75–81.
- [6] Obrst L, Chase P, Markeloff R. Developing an ontology of the cybersecurity domain. *CEUR Workshop Proc* 2012;966:49–56.
- [7] Parmelee MC. Toward an ontology architecture for cybersecurity standards. *CEUR Workshop Proc* 2010;713:116–23.
- [8] Iannacone M, Bohn S, Nakamura G, Gerth J, Huffer K, Bridges R, et al. Developing an ontology for cybersecurity knowledge graphs. In: *Proceedings of the 10th annual cyber and information security research conference*. New York:ACM, Inc.; 2015.
- [9] Jia Y, Qi Y, Shang H, et al. A Practical Approach to Constructing a Knowledge Graph for Cybersecurity[J]. *Engineering*, 2018(4)53-60.
- [10] CHEN Jianfeng, FAN Hangbo. Ontological Threat Intelligence Sharing in Cyberspace Security [J]. *Communications Technology*, 2018. 51(1):177-183.